

Postprandial metabolic events and fruit-derived phenolics: a review of the science.



There is increasing evidence that the postprandial state is an important contributing factor to chronic disease. The role of fruit phenolic compounds to protect health and lower disease risk through their actions in mitigating fed-state metabolic and oxidative stressors is of interest and the topic of the present paper. Two main questions are posed: first, what is the role of plant foods, specifically fruits rich in complex and simple phenolic compounds in postprandial metabolic management; and second, does the evidence support consuming these fruits with meals as a practical strategy to preserve health and lower risk for disease? This review provides an overview of the postprandial literature, specifically on the effect of fruits and their inherent phenolic compounds in human subjects on postprandial lipaemia, glycaemia/insulinaemia and associated events, such as oxidative stress and inflammation. Among the identified well-controlled human trials using a postprandial paradigm, >50 % of the trials used wine or wine components and the remaining used various berries. Notwithstanding the need for more research, the collected data suggest that consuming phenolic-rich fruits increases the antioxidant capacity of the blood, and when they are consumed with high fat and carbohydrate 'pro-oxidant and pro-inflammatory' meals, they may counterbalance their negative effects. Given the content and availability of fat and carbohydrate in the Western diet, regular consumption of phenolic-rich foods, particularly in conjunction with meals, appears to be a prudent strategy to maintain oxidative balance and health.